CONTROL SYSTEMS

- Shankargoud Patil

Introduction to Control Systems :

The control system is that means by which any quantity of interest in a machine, mechanism or other equipment is maintained or altered in accordance with a desired manner.

Example : The driving system of an automobile. Speed of the automobile is a function of the position of its accelerator. The desired speed can be maintained by controlling pressure on the accelerator pedal. This automobile driving system constitutes a control system.

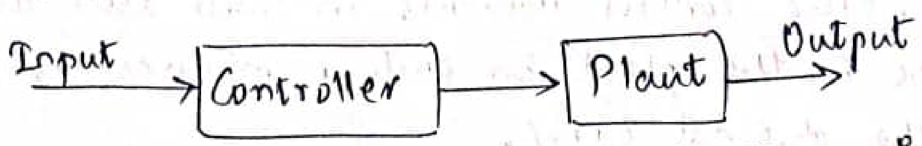
Classification of Control Systems :

Control systems can be broadly classified as,

1. Open-Loop Control system
2. Closed-Loop Control system

Open-Loop Control systems :

Any physical system which does not automatically correct for variations in its output is called as an open-loop control system.
Such a system may be represented by the block diagram

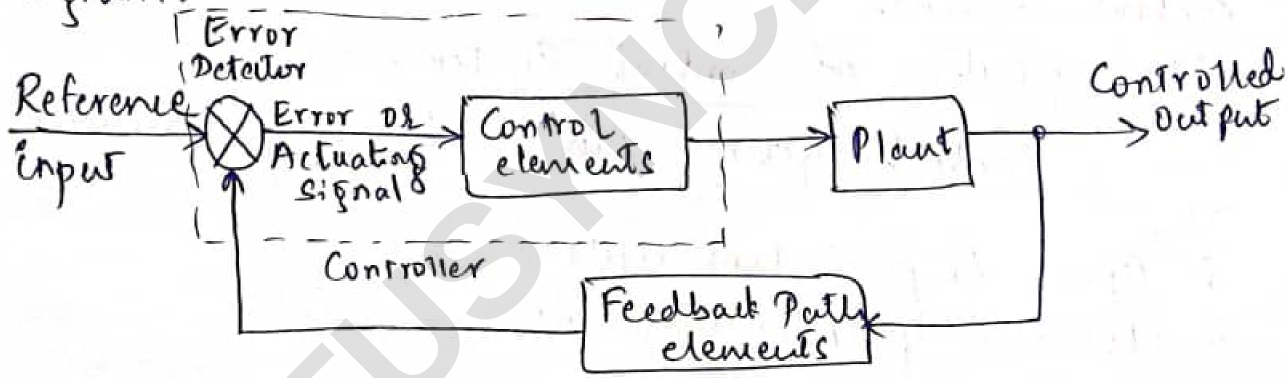


In these systems the output remains constant for a constant input signal.
Here the output is dependent on the input but input is independent of the output or changes in output.

Plant : The portion of a system which is to be controlled or regulated is called as the plant.
Controller : The element of the system itself or external to the system which controls the plant is called as controller.

Closed-Loop Control Systems :

Any physical system which automatically corrects for variations in its output is called as closed-loop control system.
It can be achieved with the help of a feedback.
Such a system may be represented by the block diagram.



An error detector compares a signal obtained through feedback elements, which is a function of the output response, with the reference input. Any difference between these two signals constitutes an error or actuating signal, which actuates the control elements. The control elements in turn alter the conditions in the plant in such a manner as to reduce the original error.

Examples :

(1) Traffic control system :

Traffic control by means of traffic signals operated on a time basis constitutes an open-loop control system. The sequence of control signals is based on a time slot given for each signal. The time slots are decided based on a traffic study. The system will not measure the density of the traffic before giving the signals. It gives the signals in sequence as per the setting irrespective of the actual traffic. Since the time slots do not change according to traffic density, the system is an open-loop control system.

This open-loop traffic control system can be made as a closed-loop system if the time-slots of the signals are based on the density of traffic. In a closed-loop traffic control system, the density of traffic is measured on all the sides and the information is fed to a computer. The timings can be dynamically changed by the computer. Such a system is an closed-loop control system.

(2) Room heating system: A room heater without any temperature sensing device is an example of an open-loop control system. It becomes a closed-loop system if a thermostat is provided.

(3) Washing Machine: A washing machine without any cleanliness measuring system is an example of an open-loop control system. In this, the soaking, washing and rinsing in the washer operate on a time basis.

It can be made a closed loop control system, if the level of cleanliness can be measured and compared with the desired cleanliness (reference input) and the difference is used to control the washing time of the machine.

Open-loop versus Closed-loop control systems

An open-loop control system can be modified as a closed-loop by providing feedback. The provision of feedback automatically corrects the changes in output due to disturbances. Hence the closed-loop control system is called as automatic control system.

Open-loop control system	Closed-loop control system
1. The open-loop control systems are simple and economical.	1. The closed-loop control systems are complex and costlier.
2. They consume less power.	2. They consume more power.
3. These are easier to construct because of less number of components required.	3. These are not easier to construct because of more number of components required.
4. Stability is not a major problem in open-loop control systems. Generally, the open-loop systems are stable.	4. Stability is a major problem in closed-loop control systems and more care is needed to design a stable closed-loop system.
5. These are inaccurate and ~~unreliab~~ unreliable	5. These are accurate and more reliable.
6. These are more sensitive to noise and other disturbances as the signal is not corrected automatically.	6. These are less sensitive to noise and other disturbances as the signal is corrected automatically.
7. The Gain is more as there is no feedback.	7. The Gain is less as there is feedback.
8. Bandwidth is less	8. Bandwidth is more.

Feedback:

Feedback can be positive or negative depending on whether it gets added up or subtracted with the reference input.

Effect of Feedback systems on Control systems Performance.

Most of the control systems use negative feedback because of its advantages. The various effects of negative feedback are

① The feedback reduces the time constant of the system. This indicates that the system becomes more fast due to feedback which improves its time response.

② The overall gain of the system gets reduced due to feedback. Thus by controlling the feedback the gain of the system can be controlled as per requirement.

③ The feedback controls the dynamics of the system by adjusting the locations of its closed loop poles. Thus the stability of the system which depends on the locations of the closed loop poles, can be controlled by using feedback.

④ The feedback increases the bandwidth i.e the frequency range over which the system performance is satisfactory.

⑤ The feedback reduces the effects of the disturbances. as it gets subtracted with the reference input.

⑥ The feedback reduces the sensitivity of the control systems to the parameter variations.

Mathematical modelling and Physical equations

The set of equations, describing the dynamic characteristics of a system is called mathematical model of the system. Here mathematical equations are the differential equations.

Mechanical systems: In mechanical systems, motion can be of different types i.e. Translational, Rotational or combination of both. The equations governing such motion in mechanical systems are often governed by Newton's Laws of motion.

Translational Motion: In mechanical systems, if motion is taking place along a straight line, such motion is translational.

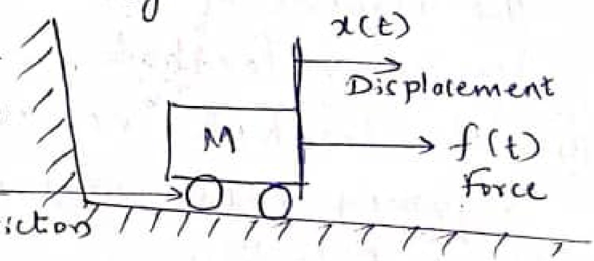
The following elements are dominantly involved in the analysis of translational motion systems.

i) Mass
ii) Spring
iii) Friction

i) Mass :

This is the property of the system itself which stores the kinetic energy of the translational motion and it has no power to store the potential energy.

The applied force $f(t)$ produces displacement $x(t)$ in the direction of the applied force and is proportional to the acceleration.



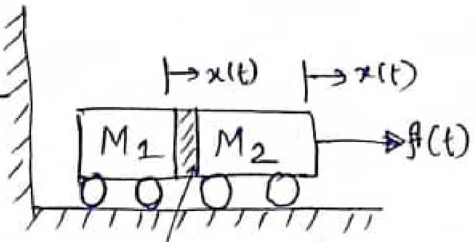
$$\therefore f(t) = M \times \text{Acceleration}.$$

$$\boxed{f(t) = M \frac{d^2}{dt^2} x(t)}$$

Taking Laplace transform and assuming zero initial conditions.

$$\boxed{F(s) = M s^2 X(s)}$$

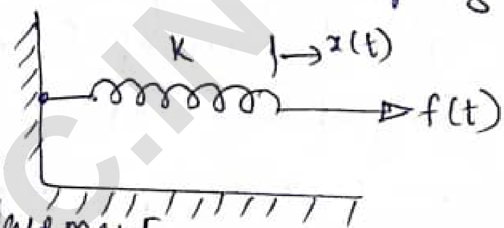
If two masses are connected together and the applied force is $f(t)$, then the mass M_1 will get displaced by $x(t)$ and mass M_2 will also displace by $x(t)$.



Rigid connection
(No friction & No Elasticity)

ii) Spring:

In actual mechanical systems, there may be an actual spring or indication of spring action because of elastic cable or belt. Spring can store the potential energy. Spring can be assumed to have linear spring constant K.



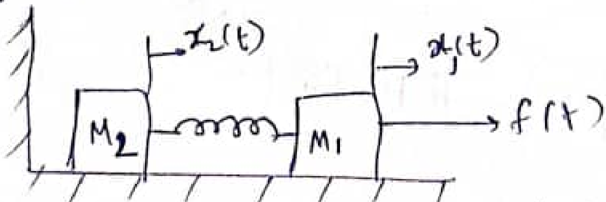
Force $f(t)$ required to cause displacement $x(t)$ in the spring is proportional to displacement.

$$\boxed{\therefore \quad f(t) = K\,x(t)}$$

Taking Laplace transform,

$$\boxed{F(s) = K\,X(s)}$$

Now consider the spring connected between two moving elements having masses M_1 and M_2 where force is applied to M_1.



Mass M_1 will get displaced by $x_1(t)$ but M_2 will get displaced by $x_2(t)$ as spring of constant K will store some potential energy and will be the cause for change in displacement.

$$\boxed{\therefore \quad F_{spring} = K\left[x_1(t) - x_2(t)\right]}$$

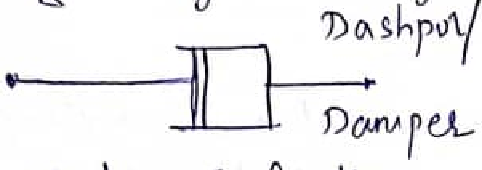
Taking Laplace Transform,

$$\boxed{F_{spring} = K\left[X_1(s) - X_2(s)\right]}$$

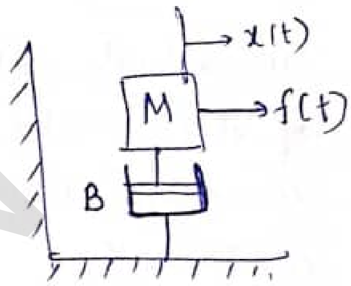
iii) Friction :

Whenever there is a motion, there exists a friction. Friction may be between moving element and fixed support or between two moving surfaces.

The friction is generally shown by a dash-pot or a damper.



Consider a mass M having friction with support, which is represented by 'B'. The friction will oppose the motion of mass M and opposing force is proportional to velocity of mass M.



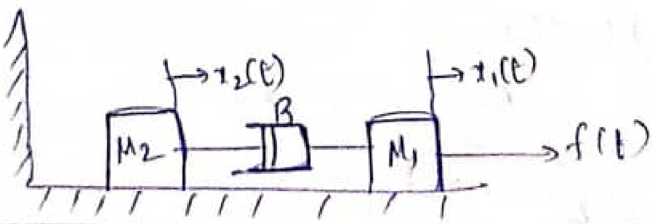
$$\therefore F_{Frictional} = B\,v(t)$$

$$\boxed{F_{Frictional} = B\frac{d}{dt}x(t)}$$

Taking Laplace Transform and assuming zero initial conditions,

$$\boxed{F_{Frictional}(s) = B\,s\,X(s)}$$

Similarly if friction is between two moving surfaces then,



$$\boxed{\therefore F_{Frictional} = B\left[\frac{d}{dt}x_1(t) - \frac{d}{dt}x_2(t)\right]}$$

Taking Laplace Transform,

$$\boxed{\therefore F_{Frictional} = Bs\left[X_1(s) - X_2(s)\right]}$$

Rotational Motion:

This is the motion about a fixed axis.

Force × distance from fixed axis is called as Torque.

The property of system which stores kinetic energy in rotational system is called Inertia and denoted by 'J' i.e moment of inertia.

Opposing Torque due to inertia 'J' is proportional to the angular acceleration (α) of that inertia

$$\boxed{T = J\frac{d^2\theta(t)}{dt^2}} \quad \text{where } \alpha = \frac{d^2\theta}{dt^2}$$

Taking Laplace Transform,

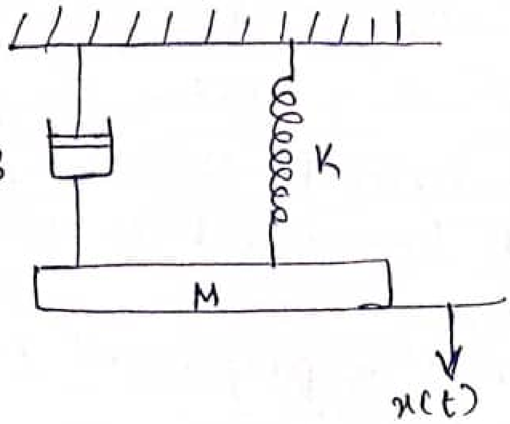
$$\boxed{T(s) = Js^2\theta(s)}$$

In rotational systems, Spring and friction behave in same manner as in Translational system.

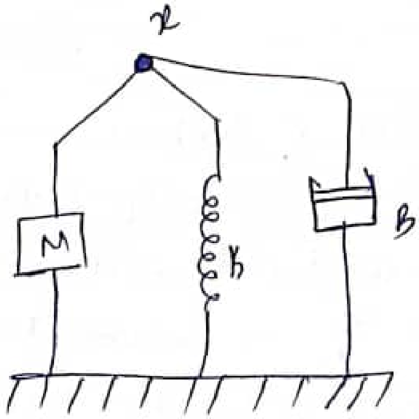
Sl.No	Translational motion	Rotational motion
1.	Mass (M)	Inertia (J)
2.	Friction (B)	Friction (B)
3.	Spring (K)	Spring (K)
4.	Force (F)	Torque (T)
5.	Displacement (x)	Angular displacement (θ)
6.	Velocity $v = \frac{dx}{dt}$	Angular velocity $\omega = \frac{d\theta}{dt}$
7.	Acceleration $= \frac{d^2x}{dt^2}$	Angular acceleration $= \frac{d^2\theta}{dt^2}$

Note: All elements which are under the influence of same displacement get connected in parallel under that node indicating the corresponding displacement.

Example 1 :

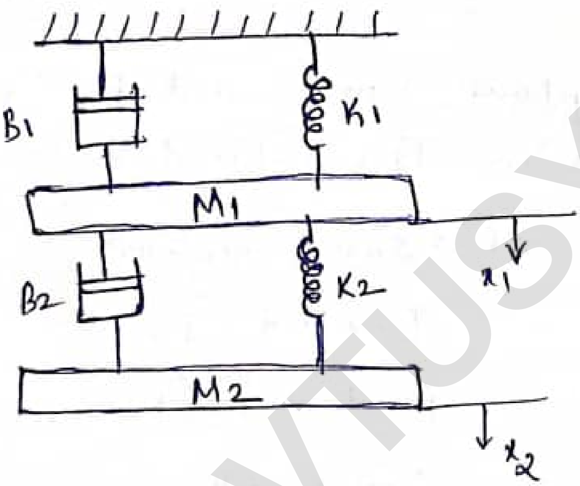


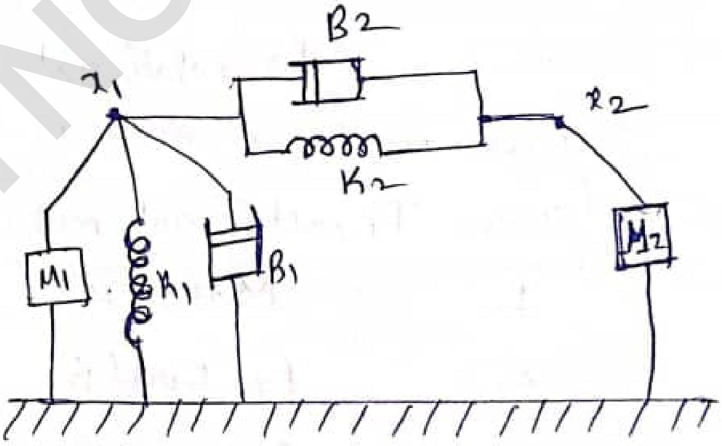
$\simeq$



Here, M, B and K all are under the influence of x(t). Hence in equivalent system all of them will get connected in parallel under the node 'x'.

Example 2 :



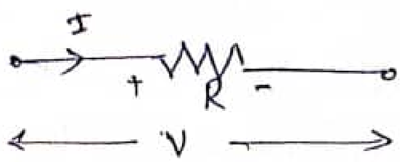


Electrical systems :

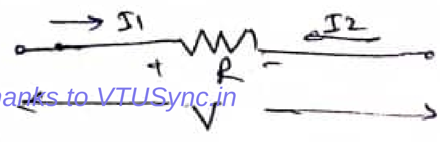
Similar to the mechanical systems, very commonly used systems are of electrical type. The behaviour such systems is governed by Ohm's law. The dominant elements of an electrical system are, (i) Resistor (ii) Inductor and (iii) Capacitor.

(i) Resistor : Consider a resistance carrying current 'I' as shown, then the voltage drop across it can be written as,

$$\boxed{V = IR}$$

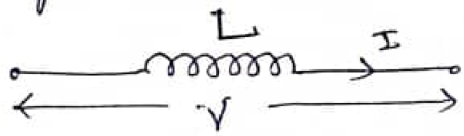


Similarly it carries a current $(I_1 - I_2)$ then for the polarity of the voltage drop shown is,



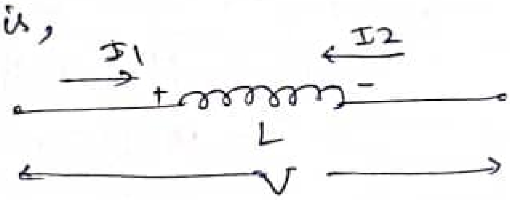
$$\boxed{V = (I_1 - I_2)R}$$

(ii) Inductor: Consider an inductor carrying current 'I' as shown, then the voltage drop across it can be written as,



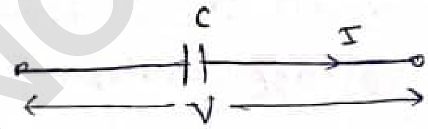
$$\boxed{V = L\frac{dI}{dt} \text{ or } I = \frac{1}{L}\int V\,dt}$$

If it carries a current $(I_1 - I_2)$ then for the polarity shown its voltage eq^n is,



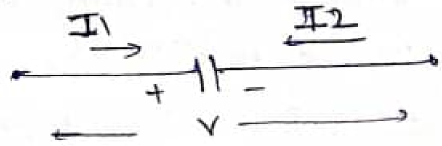
$$\boxed{V = L\frac{d}{dt}(I_1 - I_2) \text{ or } I_1 - I_2 = \frac{1}{L}\int V\,dt}$$

(iii) Capacitor: Consider a capacitor carrying current 'I' as shown, then the voltage drop across it can be written as,



$$\boxed{V = \frac{1}{C}\int I\,dt \text{ or } I = C\frac{dv}{dt}}$$

If it carries a current $(I_1 - I_2)$ then for the polarity shown its voltage eq^n is,



$$\boxed{V = \frac{1}{C}\int (I_1 - I_2)\,dt \text{ or } (I_1 - I_2) = C\frac{dv}{dt}}$$

Analogous Systems :

In between electrical and mechanical systems there exists a fixed analogy and there exists a similarity between their equilibrium equations. Due, to this, it is possible to draw an electrical system which will behave exactly similar to the given mechanical system, this is called electrical analogous of given mechanical system and vice versa.

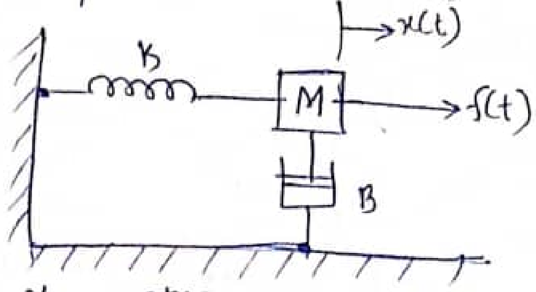
There are two methods of obtaining electrical analogous networks, namely

(i) Force- Voltage analogy.

(ii) Force- current analogy.

Mechanical System :

Consider a simple mechanical system as shown.



Due to the applied force, mass M will displace by an amount $x(t)$ in the direction of the force $f(t)$.

According to Newton's Laws of motion,

$$f(t) = Ma + Bv + Kx(t)$$

$$f(t) = M\frac{d^2}{dt^2}x(t) + B\frac{d}{dt}x(t) + K\,x(t)$$

Taking Laplace Transform,

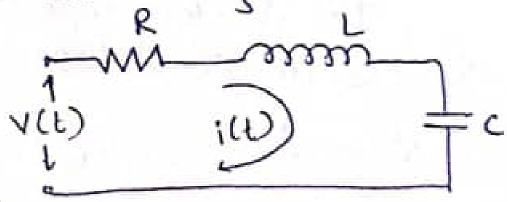
$$\boxed{F(s) = Ms^2X(s) + BsX(s) + KX(s)}$$

This is equilibrium eq^n for the given system.

Now, we will try to draw its analogous electrical Network.

Force - Voltage analogy (Loop analysis) :

In this method, to the force in mechanical system, voltage is assumed to be analogous.



The equation according to KVL can be written as,

$$v(t) = i(t)R + L\frac{d}{dt}i(t) + \frac{1}{C}\int i(t)\,dt$$

Taking Laplace Transform,

$$V(s) = R\cdot I(s) + LsI(s) + \frac{I(s)}{C\cdot s}$$

Still we can not compare $F(s)$ and $V(s)$ unless we bring them into same form,

For this we use current as rate of flow of charge

i.e $i(t) = \frac{dq}{dt}$ i.e $I(s) = sQ(s)$ or $Q(s) = \frac{I(s)}{s}$

Replacing in the above eq^n,

$$V(s) = RsQ(s) + Ls^2Q(s) + \frac{1}{C}Q(s)$$

$$\boxed{V(s) = Ls^2Q(s) + R\cdot sQ(s) + \frac{1}{C}Q(s)}$$

Comparing eq^ns for $F(s)$ and $V(s)$ it is clear that,

(i) Inductance 'L' is analogous to mass 'M'.

(ii) Resistance 'R' is analogous to friction 'B'

(iii) Reciprocal of the capacitor '$1/C$' is analogous to spring constant 'K'.

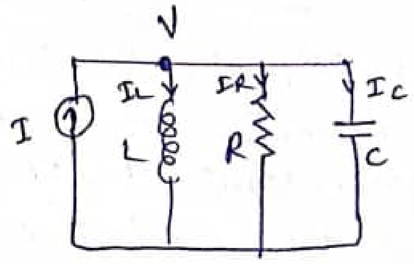
Translational	Rotational	Electrical
Force 'F'	Torque 'T'	Voltage 'V'
Mass 'M'	Inertia 'J'	Inductance 'L'
Friction 'B'	Friction 'B'	Resistance 'R'
Spring 'K'	Spring 'K'	Reciprocal of capacitor '$1/C$'
Displacement 'x'	θ	charge 'q'
velocity $\frac{dx}{dt}$	$\omega = \frac{d\theta}{dt}$	current $i = \frac{dq}{dt}$

Force-Voltage analogy.

Force-Current analogy:

In this method, to the force in mechanical system, current is assumed to be analogous.

The equation according to KCL can be written as



$$I = I_L + I_R + I_C$$

At node voltage be V,

$$I = \frac{1}{L}\int V.dt + \frac{V}{R} + C\frac{dv}{dt}$$

Taking laplace Transform,

$$I(s) = \frac{V(s)}{sL} + \frac{V(s)}{R} + s.CV(s)$$

To make this eq^n similar to that of Force, we use

$$v(t) = \frac{d\phi}{dt} \quad \text{where } \phi = \text{flux}$$

$$\therefore V(s) = s\phi(s) \quad \text{i.e } \phi(s) = \frac{V(s)}{s}$$

Substituting in the above eq^n,

$$I(s) = Cs^2\phi(s) + \frac{1}{R}s\phi(s) + \frac{1}{L}\phi(s)$$

Comparing eq^n for F(s) and I(s) it is clear that,

(i) Capacitor 'C' is analogous to Mass 'M'.

(ii) Reciprocal of resistance $1/R$ is analogous to friction 'B'

iii) Reciprocal of inductance $1/L$ is analogous to spring constant 'K'.

Translational	Rotational	Electrical
F	T	I
M	J	C
B	B	$1/R$
K	K	$1/L$
x	θ	ϕ
$\frac{dx}{dt}$	$\frac{d\theta}{dt}$	$\frac{d\phi}{dt}$

Force-Current analogy

Steps to solve problems on Analogous systems :

Step1: Identify all the displacements due to the applied force. The elements spring and friction between two moving surfaces cause change in displacement.

Step2: Draw the equivalent mechanical system based on node basis. The elements under same displacement will get connected in parallel under that node. Each displacement is represented by separate node. Element causing change in displacement (either friction or spring) is always between the two nodes.

Step 3: Write the equilibrium equations. At each node algebraic sum of all the forces acting at the node is zero.

Step 4: In F-V analogy, use following replacements and rewrite equations.

$F \to V,\ M \to L,\ B \to R,\ K \to 1/C,\ x \to q,\ \dot{x} \to i,\ x \to \int i\, dt$

Step 5: Simulate the equations using loop method. Number of displacements equal to number of loop currents.

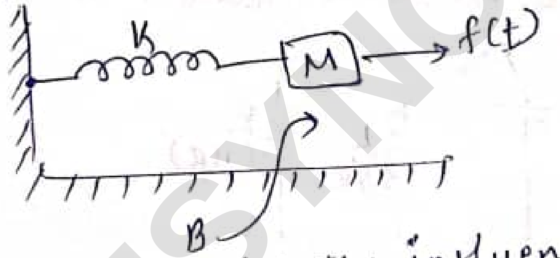
Step 6: In F-I analogy, use following replacements and rewrite equations,

$F \to I,\ M \to C,\ B \to 1/R,\ K \to 1/L,\ x \to \phi,\ \dot{x} \to e,\ x \to \int e\,dt$

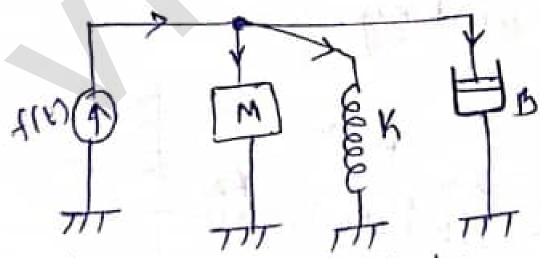
Step 7: Simulate the equations using node basis. Number of displacements equal to number of node voltages.

Numericals:

① For the physical system shown draw its equivalent system and write equilibrium equations. Hence draw its electrical analogous circuits based on
(i) Force-Voltage (ii) Force-current method.



Sol^n: All the elements are under the influence of $x(t)$, the equivalent mechanical system looks like,



The equilibrium eq^n will be,

$$f(t) = M\frac{d^2x(t)}{dt^2} + B\frac{d}{dt}x(t) + Kx(t)$$

Taking Laplace Transform,

$$F(s) = Ms^2X(s) + BsX(s) + KX(s)$$

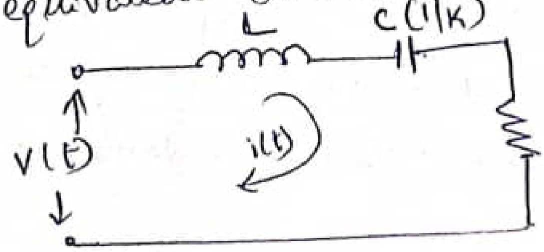
(i) Force-voltage method:

$M \to L,\ B \to R,\ K \to 1/C,\ x \to q,\ \frac{dx}{dt} \to \frac{dq}{dt} \to i,\ x \to \int i\,dt,\ \frac{d^2x}{dt^2} \to \frac{di}{dt}$

$$\therefore\ v(t) = L\frac{di}{dt} + Ri + \frac{1}{C}\int i\,dt$$

$$\therefore\ V(s) = LsI(s) + RI(s) + \frac{1}{C}\frac{I(s)}{s}$$

∴ The equivalent electrical network looks like,



Capacitor will be represented as 1/K.

(ii) **Force-current Method:**

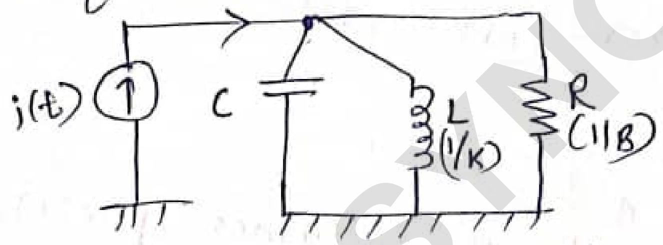
$M \to C,\ B \to \frac{1}{R},\ K \to \frac{1}{L},\ x \to \phi,\ \frac{dx}{dt} \to \frac{d\phi}{dt} \to V(t),$

$x \to \int V(t)\,dt,\ \frac{d^2x}{dt^2} \to \frac{dV(t)}{dt}$

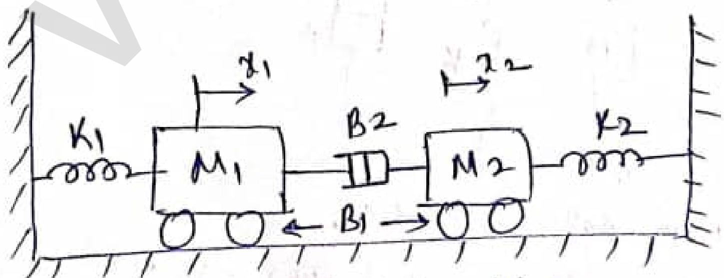
$$\therefore\ i(t) = C\frac{dV(t)}{dt} + \frac{1}{L}\int V(t)\,dt + \frac{1}{R}V(t)$$

$$I(s) = C \cdot sV(s) + \frac{1}{Ls}V(s) + \frac{1}{R}V(s)$$

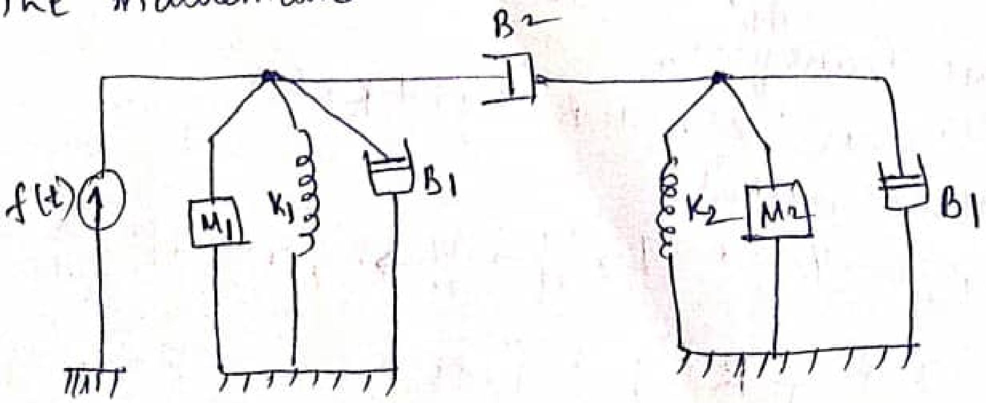
∴ The equivalent electrical network looks like,



② Construct mathematical model for the mechanical system shown. Then draw its electrical equivalent circuit based on Force-Voltage analogy.



Soln: The mathematical model looks like,



The equilibrium equations at two nodes are,

$$f(t) = M_1 \frac{d^2 x_1}{dt^2} + B_1 \frac{dx_1}{dt} + K_1 x_1 + B_2 \frac{d}{dt}(x_1 - x_2)$$

$$0 = B_2 \frac{d(x_2 - x_1)}{dt} + M_2 \frac{d^2 x_2}{dt^2} + B_1 \frac{dx_2}{dt} + K_2 x_2$$

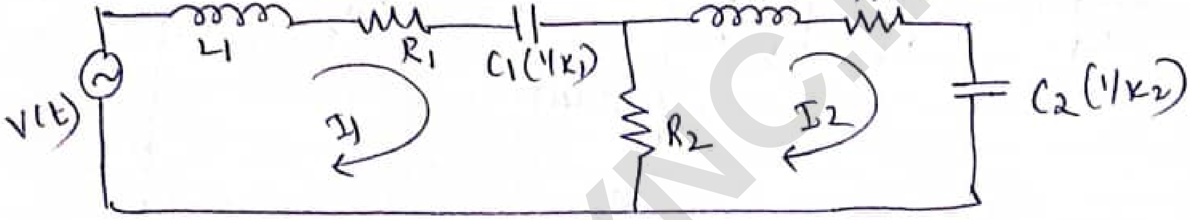
Force-Voltage analogy:

$M \to L$, $B \to R$, $K \to 1/C$, $x \to q$, $\frac{dx}{dt} \to i$, $x \to \int i\,dt$, $\frac{d^2x}{dt^2} \to \frac{di}{dt}$

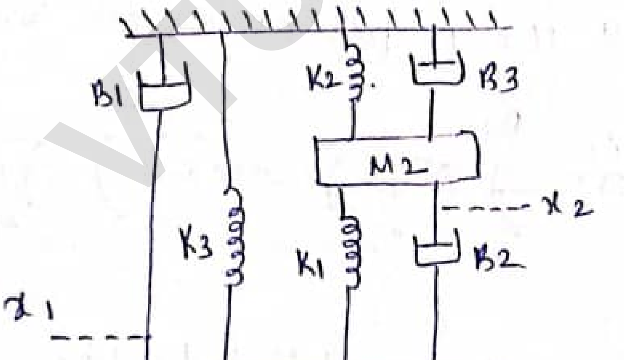
$$\therefore v(t) = L_1 \frac{di_1}{dt} + R_1 i + \frac{1}{C_1}\int i\,dt + R_2 (i_1 - i_2)$$

$$0 = B_2 (i_2 - i_1) + L_2 \frac{di_2}{dt} + R_1 i_2 + \frac{1}{C_2}\int i_2\,dt$$

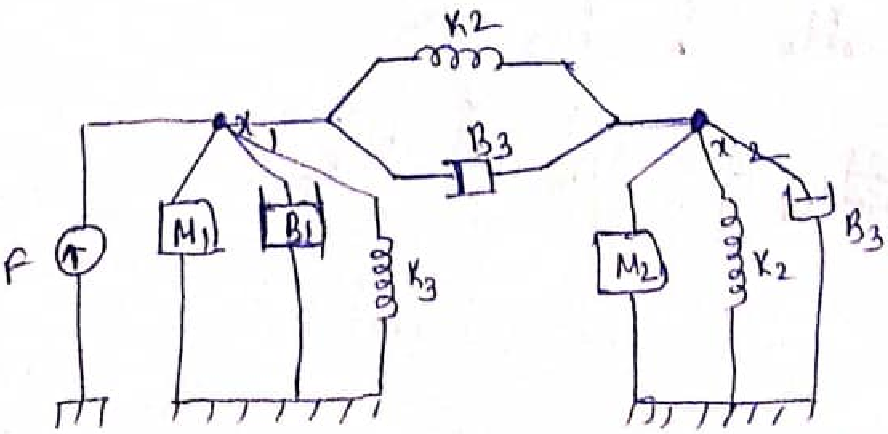
Using these two eqns the network can be drawn as,



(3) For a mechanical system shown below, obtain force-voltage analogous electrical network.



Soln:



Mathematical model equivalent

The equilibrium eqns are,

$$F = M_1\frac{d^2x_1}{dt^2} + B_1\frac{dx_1}{dt} + K_3x_1 + K_1(x_1-x_2) + B_2\frac{d(x_1-x_2)}{dt}$$

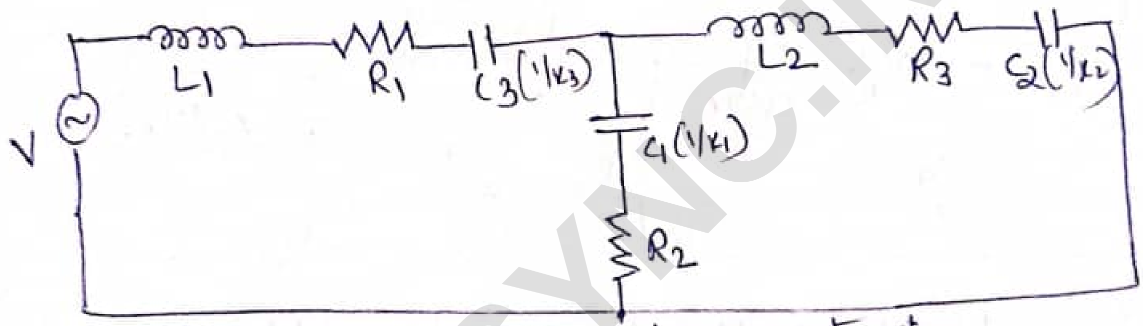
$$0 = M_2\frac{d^2x_2}{dt^2} + B_3\frac{dx_2}{dt} + K_2x_2 + K_1(x_2-x_1) + B_2\frac{d}{dt}(x_2-x_1)$$

Using Force-Voltage analogy,

$M \to L$, $B \to R$, $K \to 1/C$, $\frac{dx}{dt} \to i$, $x \to \int i\,dt$, $\frac{d^2x}{dt^2} \to \frac{di}{dt}$

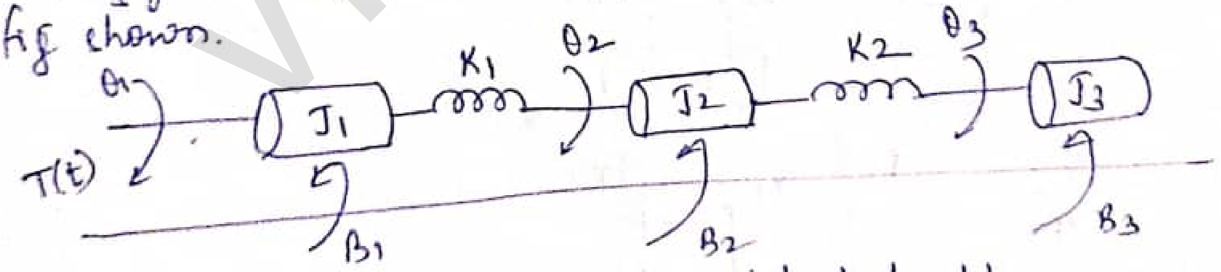
$$\therefore V = L_1\frac{di_1}{dt} + R_1 \frac{d}{dt} i_1 + \frac{1}{C_3}\int i\,dt + \frac{1}{C_1}\int(i_1-i_2)dt + R_2(i_1-i_2)$$

$$0 = L_2\frac{di_2}{dt} + R_3 i_2 + \frac{1}{C_2}\int i_2\,dt + \frac{1}{C_1}\int(i_2-i_1)dt + R_2(i_2-i_1)$$

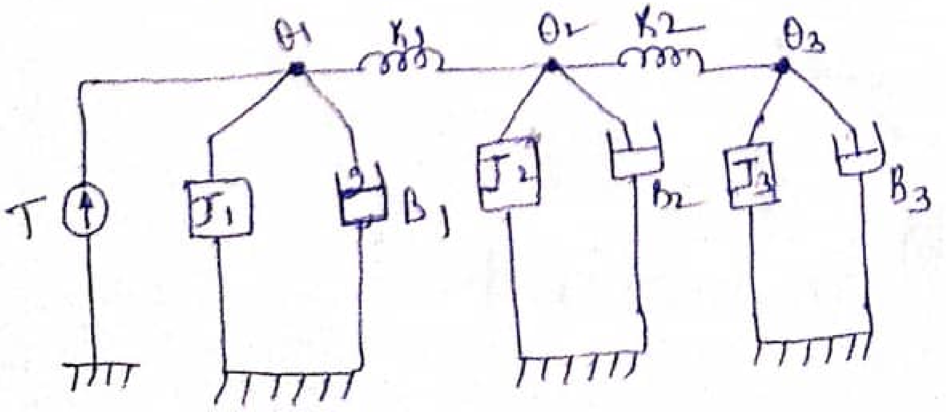


Force-Voltage Electrical analogous network.

(4) Draw the electrical network based on Torque-current analogy and give all the performance eqns for the fig shown.



Soln: The equivalent Mathematical model looks like,



The equilibrium eq^{ns} are,

$$T = J_1 \frac{d^2\theta_1}{dt^2} + B_1 \frac{d}{dt}\theta_1 + K_1(\theta_1 - \theta_2)$$

$$0 = J_2 \frac{d^2\theta_2}{dt^2} + B_2 \frac{d}{dt}\theta_2 + K_1(\theta_2 - \theta_1) + K_2(\theta_2 - \theta_3)$$

$$0 = J_3 \frac{d^2\theta_3}{dt^2} + B_3 \frac{d\theta_3}{dt} + K_2(\theta_3 - \theta_2)$$

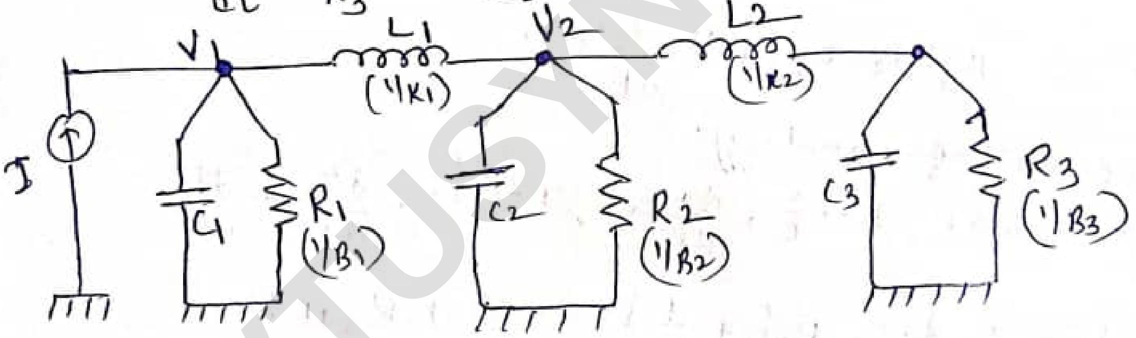
Using Torque- Current analogy,

$J \to C$, $B \to \frac{1}{R}$, $K_1 \to \frac{1}{L}$, $\frac{d\theta}{dt} \to V$, $\theta \to \int V dt$, $\frac{d^2\theta}{dt^2} \to \frac{dV}{dt}$

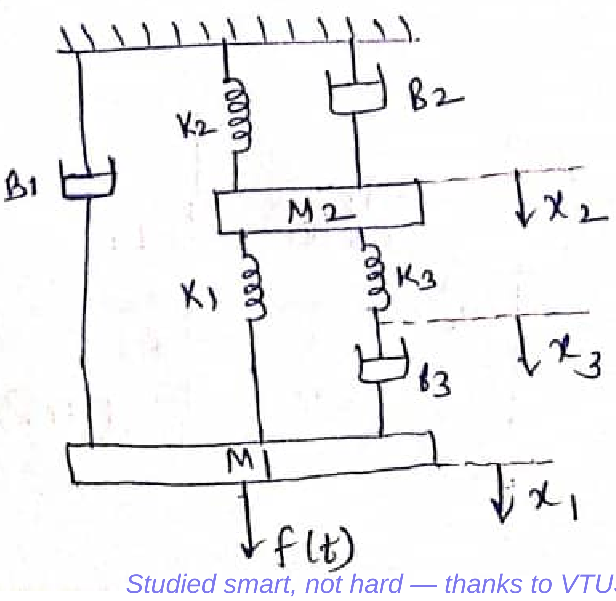
$$I = C_1 \frac{dv}{dt} + \frac{1}{R_1} V_1 + \frac{1}{L_2} \int (V_1 - V_2)\, dt$$

$$0 = C_2 \frac{dv}{dt} + \frac{1}{R_2} V_2 + \frac{1}{L_1} \int (V_2 - V_1)\, dt + \frac{1}{L_2} \int (V_2 - V_3)\, dt$$

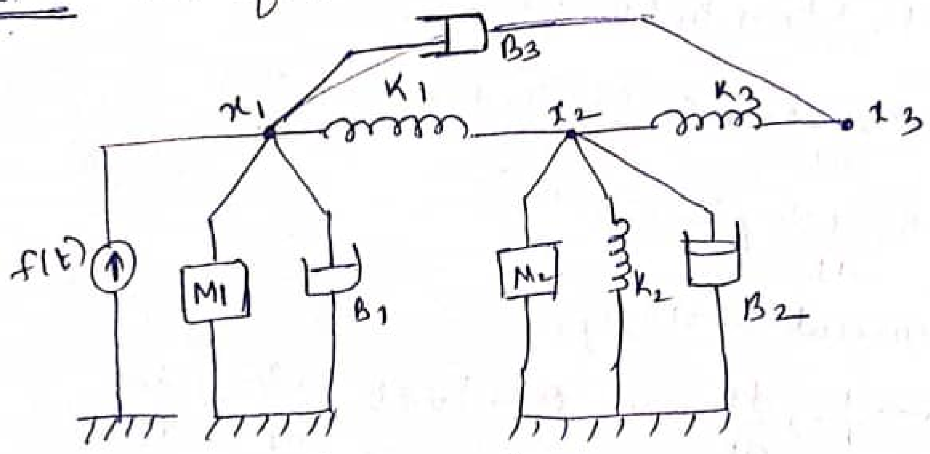
$$0 = C_3 \frac{dv}{dt} + \frac{1}{R_3} V_3 + \frac{1}{L_3} \int (V_3 - V_2)\, dt$$



(5) Draw the Force- Voltage and Force- Current analogous circuits for the mechanical systems shown below with necessary equations.



Sol^n: The equivalent mathematical model looks like,



The equilibrium eq^n are,

$$f(t) = M_1 \frac{d^2 x_1}{dt^2} + B_1 \frac{dx_1}{dt} + K_1(x_1 - x_2) + B_3 \frac{d(x_1 - x_3)}{dt}$$

$$0 = M_2 \frac{d^2 x_2}{dt^2} + B_2 \frac{dx_2}{dt} + K_2 x_2 + K_1(x_2 - x_1) + K_3(x_2 - x_3)$$

$$0 = K_3(x_3 - x_2) + B_3 \frac{d(x_3 - x_1)}{dt}$$

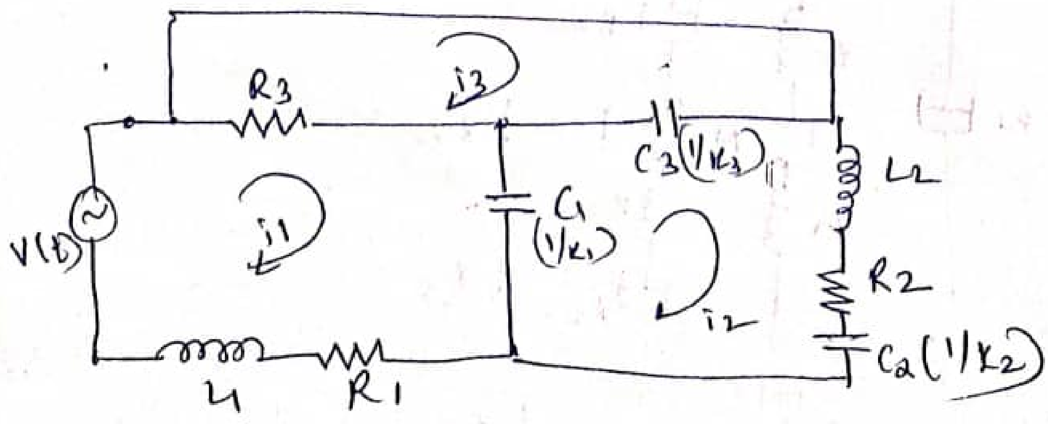
Force-Voltage analogy:

$M \to L,\ B \to R,\ K \to \frac{1}{C},\ \frac{dx}{dt} \to i,\ x \to \int i,\ \frac{d^2x}{dt^2} \to \frac{di}{dt}$

$$\therefore V(t) = L_1 \frac{di_1}{dt} + R_1 i_1 + \frac{1}{C_1}\int (i_1 - i_2)\,dt + R_3(i_1 - i_3)$$

$$0 = L_2 \frac{di_1}{dt} + R_2 i_2 + \frac{1}{C_2}\int i_2\,dt + \frac{1}{C_1}\int (i_2 - i_1)\,dt + \frac{1}{C_3}\int (i_2 - i_3)$$

$$0 = \frac{1}{C_3}\int (i_3 - i_2)\,dt + R_3(i_3 - i_1)$$



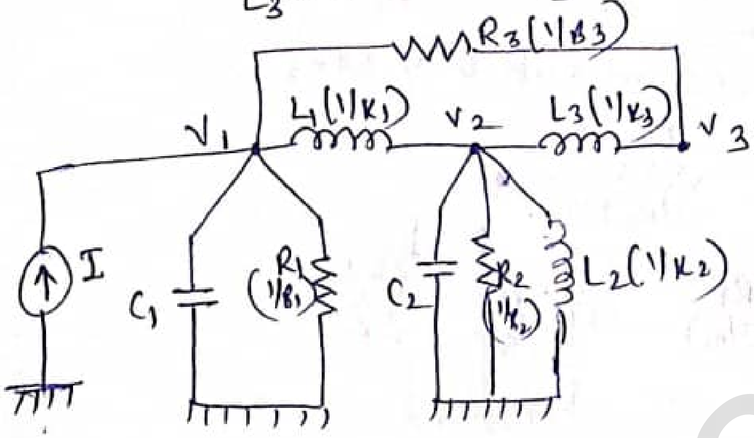
Force - current analogy:

$M \to C,\ B \to \frac{1}{R},\ K \to \frac{1}{L},\ \frac{dx}{dt} \to V,\ x \to \int V \cdot dt,\ \frac{d^2x}{dt} \to \frac{dV}{dt}$

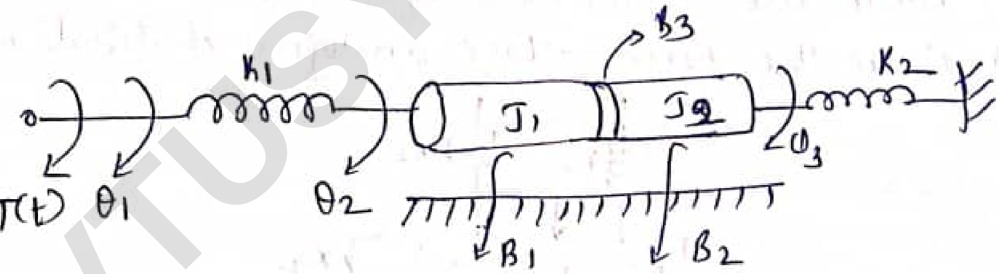
$$\therefore I(t) = C_1 \frac{dV_1}{dt} + \frac{1}{R_1} V_1 + \frac{1}{L_1} \int (V_1 - V_2)\,dt + \frac{1}{R_3}(V_1 - V_3)$$

$$0 = C_2 \frac{dV_2}{dt} + \frac{1}{R_2} V_2 + \frac{1}{L_2} \int V_2\,dt + \frac{1}{L_1} \int (V_2 - V_1)\,dt + \frac{1}{L_3} \int (V_2 - V_3)\,dt$$

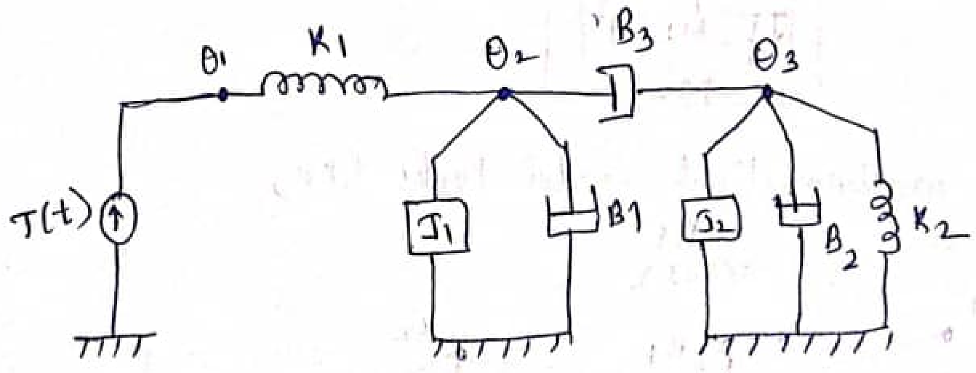
$$0 = \frac{1}{L_3} \int (V_3 - V_2)\,dt + \frac{1}{R_3}(V_3 - V_1)$$



(6) For the mechanical model shown draw its mathematical model and obtain electrical analogous based on force-current analogy.



Sol: The mathematical model looks like,



The equilibrium eqns at three nodes are,

$$T(t) = K_1(\theta_1 - \theta_2)$$

$$0 = J_1 \frac{d^2\theta_2}{dt^2} + B_1 \frac{d\theta_2}{dt} + K_1(\theta_2 - \theta_1) + B_3 \frac{d(\theta_2 - \theta_3)}{dt}$$

$$0 = J_2 \frac{d^2\theta_3}{dt^2} + B_2 \frac{d\theta_3}{dt} + K_2\theta_3 + B_3 \frac{d(\theta_3 - \theta_2)}{dt}$$

Using Force-current analogy:

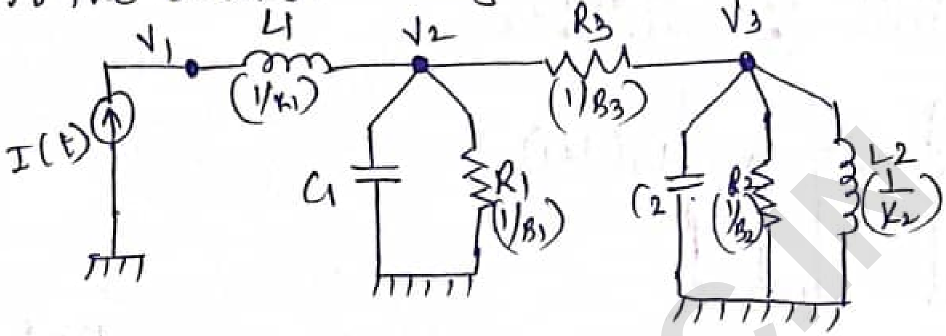
$J \to C$, $B \to \frac{1}{R}$, $K \to \frac{1}{L}$, $\frac{d\theta}{dt} \to V$, $\theta \to \int V\,dt +$ $\frac{d^2\theta}{dt^2} \Rightarrow \frac{dV}{dt}$

$$\therefore \quad I(t) = \frac{1}{L_1}\int (V_1 - V_2)\,dt$$

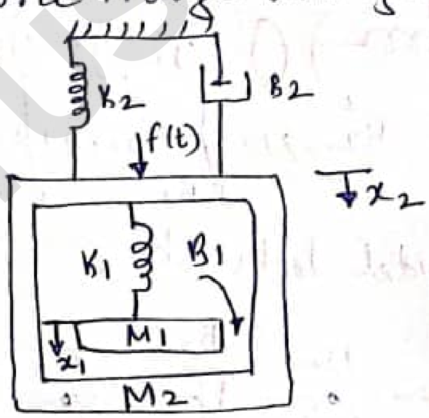
$$0 = C_1\frac{dV_2}{dt} + \frac{1}{R_1}V_2 + \frac{1}{L_1}\int (V_2 - V_1)\,dt + \frac{1}{R_3}(V_2 - V_3)$$

$$0 = C_2\frac{dV_3}{dt} + \frac{1}{R_2}V_3 + \frac{1}{L_2}\int V_3\,dt + \frac{1}{R_3}(V_3 - V_2)$$

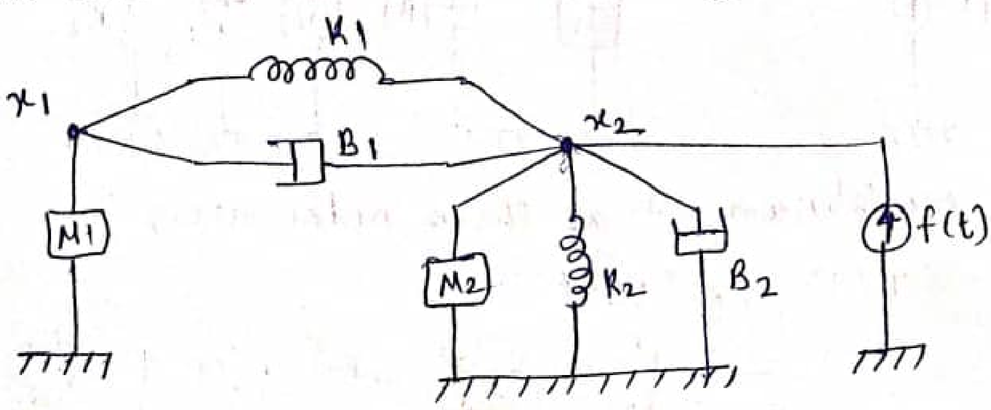
$\therefore$ The electrical analogous network looks like,



(7) For the mechanical system shown,
(i) Draw the mathematical model
(ii) Write the differential equations
(iii) Draw the force-voltage analogous electrical network.



Sol^n: (i) The mathematical model looks like,



(ii) The differential equations can be written as,

$$0 = M_1 \frac{d^2 x_1}{dt^2} + K_1(x_1 - x_2) + B_1 \frac{d(x_1 - x_2)}{dt}$$

$$f(t) = M_2 \frac{d^2 x_2}{dt^2} + K_2 x_2 + B_2 \frac{dx_2}{dt} + K_1(x_2 - x_1) + B_1 \frac{d(x_2 - x_1)}{dt}$$

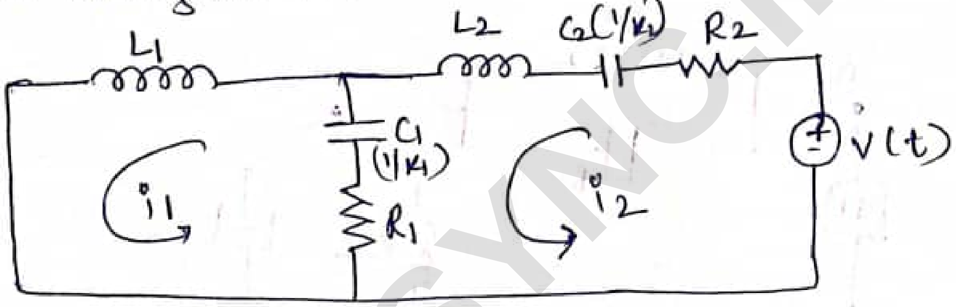
(iii) Making use of Force-voltage analogy,

$F \to V$, $M \to L$, $B \to R$, $K \to \frac{1}{C}$, $\frac{dx}{dt} \to i$, $x \to \int i\,dt$, $\frac{d^2x}{dt^2} \to \frac{di}{dt}$

$$0 = L_1 \frac{di_1}{dt} + \frac{1}{C_1} \int (i_1 - i_2) + R_1 (i_1 - i_2)$$

$$V(t) = L_2 \frac{di_2}{dt} + \frac{1}{C_2} \int i_2\,dt + R_2 i_2 + \frac{1}{C_1} \int (i_2 - i_1)\,dt + R_1 (i_2 - i_1)$$

The analogous electrical network will be,



Transfer Function :

It is defined as the ratio of Laplace transform of output of the system to the Laplace transform of input of the system, under the assumptions that all initial conditions are zero.

Symbolically it can be written as,

$r(t) \to$ input
$c(t) \to$ output

$R(s) \to$ Laplace transform of input
$C(s) \to$ Laplace transform of output
$T(s) \to$ Transfer function

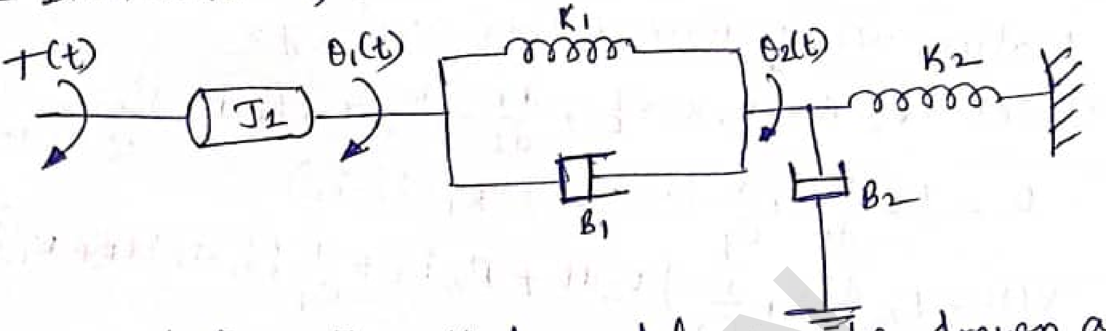
$$\boxed{\therefore \text{Transfer function } T(s) = \frac{C(s)}{R(s)}}$$

⑧ For the system shown, find the transfer function,

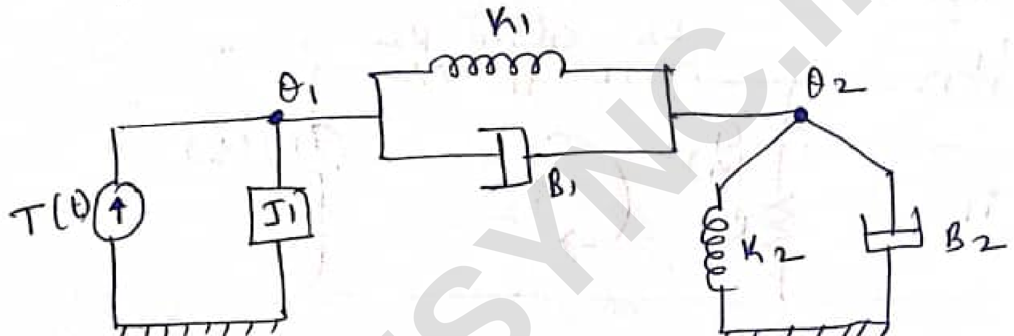
$$G(S) = \frac{\theta_2(s)}{T(s)}$$

Consider $J_1 = 1\ kgm^2$, $K_1 = 1\ Nm/rad$, $K_2 = 1\ Nm/rad$

$B_1 = 1\ Nm/rad/sec$, $B_2 = 1\ Nm/rad/sec$.



Sol: The equivalent mathematical model can be drawn as,



The equilibrium eqns can be written as,

$$T(t) = J_1 \frac{d^2\theta_1}{dt^2} + B_1 \frac{d}{dt}(\theta_1 - \theta_2) + K_1(\theta_1 - \theta_2)$$

$$0 = B_2 \frac{d}{dt}\theta_2 + K_2\theta_2 + K_1(\theta_2 - \theta_1) + B_1 \frac{d}{dt}(\theta_2 - \theta_1)$$

Taking Laplace Transform of both the eqns,

$$T(s) = s^2 J_1 \theta_1(s) + B_1 s[\theta_1(s) - \theta_2(s)] + K_1[\theta_1(s) - \theta_2(s)]$$

$$0 = B_2 s \cdot \theta_2(s) + K_2\theta_2(s) + K_1[\theta_2(s) - \theta_1(s)] + B_1 s[\theta_2(s) - \theta_1(s)]$$

Rearranging the above equations,

$$T(s) = [s^2 J_1 + sB_1 + K_1]\theta_1(s) - [sB_1 + K_1]\theta_2(s) \rightarrow ①$$

$$0 = -[K_1 + sB_1]\theta_1(s) + [s(B_1+B_2) + (K_1+K_2)]\theta_2(s) \rightarrow ②$$

$$\therefore \theta_1(s) = \frac{s(B_1+B_2) + (K_1+K_2)}{(sB_1+K_1)} \cdot \theta_2(s)$$

Substituting this in eqn ① we get,

$$\therefore T(s) = \frac{[s^2J_1 + sB_1 + K_1][s(B_1+B_2) + (K_1+K_2)]\theta_2(s)}{(sB_1+K_1)} - (sB_1+K_1)\theta_2(s)$$

Using the given values,

$$T(s) = \frac{(s^2+s+1)(2s+2)}{(s+1)}\theta_2(s) - (s+1)\theta_2(s)$$

$$\therefore \frac{\theta_2(s)}{T(s)} = \left[\frac{(s^2+s+1)2\cancel{(s+1)}}{\cancel{(s+1)}} - (s+1)\right]\theta_2(s)$$

$$= (2s^2+2s+2-s-1)\,\theta_2(s)$$

$$T(s) = (2s^2+s+1)\,\theta_2(s)$$

$\therefore$ Transfer function, $\boxed{\frac{\theta_2(s)}{T(s)} = \frac{1}{2s^2+s+1}}$